



Software Requirements Specification

For AUCDT/Hubtel Payment Integration

AsanSka University College of Design and Technology

August 2025

Contents

1	Introduction	2
1.1	Purpose	2
1.2	Scope	2
1.3	Definitions, Acronyms, and Abbreviations	2
1.4	References	3
1.5	Overview	3
2	Overall Description	3
2.1	Product Perspective	3
2.2	Product Functions	4
2.3	User Classes and Characteristics	4
2.3.1	Primary Users	4
2.3.2	Secondary Users	4
2.4	Operating Environment	5
2.5	Design and Implementation Constraints	5
2.6	Assumptions and Dependencies	6
3	Specific Requirements	6
3.1	Functional Requirements	6
3.1.1	FR-001: User Registration	6
3.1.2	FR-002: Account Verification	6
3.1.3	FR-003: Tuition Fee Calculation	7
3.1.4	FR-004: Multi-Gateway Payment Processing	7
3.1.5	FR-005: Comprehensive Application Management	7
4	System Features	7
4.1	User Registration and Verification	7
4.1.1	Description	7
4.1.2	Functional Requirements	8
4.1.3	Priority	8
4.2	Payment Integration and Financial Management	8
4.2.1	Description	8
4.2.2	Functional Requirements	8
4.2.3	Priority	8
4.3	Application Management	8
4.3.1	Description	8
4.3.2	Functional Requirements	9
4.3.3	Priority	9
5	External Interface Requirements	9
5.1	User Interfaces	9
5.2	Hardware Interfaces	9
5.3	Software Interfaces	9
5.4	Communication Interfaces	10

6 Non-Functional Requirements	10
6.1 Performance Requirements	10
6.2 Security Requirements	10
6.3 Reliability Requirements	10
6.4 Availability Requirements	11
6.5 Maintainability Requirements	11
6.6 Portability Requirements	11
7 Appendices	11
7.1 Database Schema	11
7.2 API Endpoints	12
7.3 User Journey Summary	13
7.4 Implementation Timeline	14
7.5 Compliance Checklist	14
Document Control	14

1. Introduction

1.1. Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the AsanSka University College of Design and Technology Online Application Portal with integrated payment processing via Stripe and PayPal. The system is designed to modernise the application process for prospective students whilst ensuring secure payment processing and compliance with UK data protection regulations.

1.2. Scope

The AsanSka University College Application Portal is a web-based platform that facilitates:

- Streamlined student registration with UCAS integration capabilities
- Dynamic fee calculation based on course type and student residency status (Home/EU vs International)
- Secure payment processing through Stripe and PayPal integration
- Comprehensive application data collection aligned with UCAS standards for creative and design programmes
- Administrative oversight and compliance reporting
- Integration with existing Student Information Systems (SIS)
- Portfolio upload and assessment tools specific to design and technology programmes

The system will benefit prospective students by providing an intuitive application journey tailored to creative disciplines and benefit University College staff by automating administrative processes whilst maintaining rigorous data governance standards.

1.3. Definitions, Acronyms, and Abbreviations

AUCDT	AsanSka University College of Design and Technology
UCAS	Universities and Colleges Admissions Service
SIS	Student Information System
GDPR	General Data protection Regulation
API	Application Programming Interface
SRS	Software Requirements Specification
TDD	Test-Driven Development
UI	User Interface

SMS	Short Message Service
PCI DSS	Payment Card Industry Data Security Standard
ICO	Information Commissioner's Office

1.4. References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- Stripe Payment Gateway API Documentation v2023
- PayPal Developer API Documentation
- GDPR Compliance Guidelines (ICO)
- UCAS Technical Integration Standards
- AsanSka University College Data protection Policy
- Java Spring Boot Framework Documentation
- Angular Framework Documentation

1.5. Overview

This SRS is organized into sections covering overall system description, specific functional requirements, system features, interface requirements, and non-functional requirements. The document serves as the primary reference for development, testing, and validation activities.

2. Overall Description

2.1. Product Perspective

The AsanSka University College Application Portal is a comprehensive web application that integrates with:

- Stripe and PayPal payment gateways for secure transaction processing
- Email and SMS services for applicant verification and communication
- PostgreSQL database with encryption at rest
- UCAS Connect API for application data exchange
- Existing University College SIS for student record management
- Microsoft 365 for email services integration
- Creative portfolio management systems for design and technology submissions
- Web browsers across all major platforms with accessibility compliance

2.2. Product Functions

The primary functions include:

- Secure applicant registration with enhanced verification processes
- Account verification via email and mobile authentication
- Dynamic tuition fee calculation based on residency status and programme selection
- Multi-gateway payment processing with PCI DSS compliance
- Comprehensive application data collection following UCAS standards for creative programmes
- Portfolio upload and management system for design and technology applications
- Document upload and verification system
- Administrative dashboard with comprehensive reporting
- Integration with University College's existing student management systems

2.3. User Classes and Characteristics

2.3.1. Primary Users

Prospective Applicants Individuals seeking admission to AsanSka University College design and technology programmes

Technical expertise Basic to intermediate digital literacy with creative software familiarity

Frequency of use Primarily single-use during application period

Security level Authenticated applicant access with role-based permissions

2.3.2. Secondary Users

Academic Staff Programme coordinators, design tutors, and admissions staff

Technical expertise Intermediate digital skills with training provided

Frequency of use Regular during admissions cycles

Security level Academic staff access with programme-specific permissions

Admissions Office Professional services staff managing creative programme applications

Technical expertise Advanced administrative system knowledge

Frequency of use Daily operational use

Security level Administrative access with full application oversight

IT Support Technical staff maintaining system integrity

Technical expertise Expert-level system administration

Frequency of use Ongoing maintenance and support

Security level System administrator privileges

2.4. Operating Environment

- Server Environment: AWS Cloud Infrastructure or University-managed servers
- Database: PostgreSQL 13 or higher with encryption capabilities
- Backend Framework: Java Spring Boot with Spring Security
- Frontend Framework: Angular with Material Design
- Web Servers: Nginx with SSL/TLS termination
- Client Environment: Modern web browsers supporting HTML5, CSS3, and ES6+ (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- Network: JANET connectivity with internet failover
- Mobile Compatibility: Responsive design supporting iOS Safari and Android Chrome

2.5. Design and Implementation Constraints

- Compliance with GDPR and UK Data protection Act 2018
- Integration with Stripe and PayPal payment gateways
- Adherence to WCAG 2.1 AA accessibility standards
- PCI DSS compliance for payment processing
- Integration with existing University College authentication systems (Active Directory/SAML)
- Performance requirements supporting peak application periods
- Data residency requirements within UK/EU jurisdictions
- Support for large file uploads for creative portfolios and design work
- Integration with creative software file formats (Adobe Creative Suite, CAD files, etc.)

2.6. Assumptions and Dependencies

- Stripe and PayPal payment gateway availability and reliability
- JANET network connectivity with appropriate bandwidth for large file transfers
- Microsoft 365 integration for email and authentication services
- UCAS API availability for data exchange
- User device compatibility with contemporary web standards and creative file formats
- Compliance with evolving UK data protection legislation
- Integration capabilities with existing University College systems
- Cloud storage solutions for portfolio and creative work submissions

3. Specific Requirements

3.1. Functional Requirements

3.1.1. FR-001: User Registration

Priority High

Description The system shall allow prospective applicants to create accounts with minimal required information.

Inputs First name, last name, email address, phone number

Processing Validate input data, create user account, generate verification token

Outputs User account created, verification token sent via email/SMS

3.1.2. FR-002: Account Verification

Priority High

Description The system shall verify user accounts through token-based verification.

Inputs Verification token

Processing Validate token, check expiry, activate user account

Outputs Account activation confirmation

3.1.3. FR-003: Tuition Fee Calculation

Priority	High
Description	The system shall dynamically calculate application and tuition fees based on programme selection and student residency status.
Inputs	Programme code, student residency status (Home/EU/International), application type
Processing	Determine fee structure, apply any applicable discounts or scholarships, calculate total fees
Outputs	Detailed fee breakdown, payment options, estimated total cost

3.1.4. FR-004: Multi-Gateway Payment Processing

Priority	High
Description	The system shall integrate with multiple payment providers (Stripe, PayPal) for secure transaction processing.
Inputs	Payment request, applicant information, fee amount, preferred payment method
Processing	Route to appropriate payment gateway, process transaction, handle callbacks and webhooks
Outputs	Payment confirmation, transaction reference, receipt generation

3.1.5. FR-005: Comprehensive Application Management

Priority	High
Description	The system shall collect and manage comprehensive application information in accordance with UCAS standards and University College requirements, with special provisions for creative portfolio submissions.
Inputs	Personal details, educational qualifications, references, personal statement, creative portfolio, supporting documentation
Processing	Validate application completeness, perform eligibility checks, process creative submissions, store securely with audit trail
Outputs	Complete application profile, application reference number, status tracking, portfolio review interface

4. System Features

4.1. User Registration and Verification

4.1.1. Description

Streamlined registration process requiring only essential information followed by secure account verification.

4.1.2. Functional Requirements

- REQ-4.1.1: System shall validate email format and phone number format
- REQ-4.1.2: System shall generate unique verification tokens with expiration times
- REQ-4.1.3: System shall send verification tokens via email and SMS
- REQ-4.1.4: System shall prevent duplicate registrations for the same email/phone

4.1.3. Priority

High – Critical for user onboarding

4.2. Payment Integration and Financial Management

4.2.1. Description

Robust payment processing through multiple gateways with comprehensive financial reporting and compliance.

4.2.2. Functional Requirements

- REQ-4.2.1: System shall determine fee structure based on programme type and residency status
- REQ-4.2.2: System shall provide transparent fee breakdown before payment authorisation
- REQ-4.2.3: System shall securely integrate with Stripe for card payments and direct debit
- REQ-4.2.4: System shall integrate with PayPal for alternative payment methods
- REQ-4.2.5: System shall maintain PCI DSS compliance throughout payment processing
- REQ-4.2.6: System shall generate VAT-compliant receipts and invoices
- REQ-4.2.7: System shall support instalment payment plans where applicable

4.2.3. Priority

High – Critical for revenue management and regulatory compliance

4.3. Application Management

4.3.1. Description

Comprehensive application data collection and management capabilities.

4.3.2. Functional Requirements

- REQ-4.3.1: System shall provide forms for educational background entry
- REQ-4.3.2: System shall collect family and personal information
- REQ-4.3.3: System shall validate required fields and data formats
- REQ-4.3.4: System shall allow application data modification before submission
- REQ-4.3.5: System shall generate application reference numbers

4.3.3. Priority

Medium – Important for complete application processing

5. External Interface Requirements

5.1. User Interfaces

- UI-001: Responsive web design supporting desktop and mobile devices
- UI-002: Intuitive navigation with clear progress indicators
- UI-003: Accessible design following WCAG 2.1 guidelines
- UI-004: Multi-language support (English primary)

5.2. Hardware Interfaces

- HI-001: Standard web server hardware requirements
- HI-002: Database server with adequate storage and processing capacity

5.3. Software Interfaces

- SI-001: Stripe Payment Gateway API v2023 with webhook integration
- SI-002: PayPal REST API v2 for alternative payment processing
- SI-003: Microsoft 365 API for email services and authentication
- SI-004: UCAS Connect API for application data exchange
- SI-005: PostgreSQL database with encrypted connections
- SI-006: University SIS integration via secure REST APIs
- SI-007: SMS service provider API (Twilio or equivalent UK provider)

5.4. Communication Interfaces

- CI-001: HTTPS protocol for all client-server communication
- CI-002: RESTful API design for frontend-backend communication
- CI-003: Secure webhook endpoints for payment callbacks

6. Non-Functional Requirements

6.1. Performance Requirements

- PERF-001: System shall support 100 concurrent users
- PERF-002: Page load times shall not exceed 3 seconds
- PERF-003: Payment processing shall complete within 30 seconds
- PERF-004: Database queries shall execute within 2 seconds

6.2. Security Requirements

- SEC-001: All data transmission shall use TLS 1.3 encryption as minimum standard
- SEC-002: User passwords shall be hashed using bcrypt with appropriate salt rounds
- SEC-003: Payment information shall be processed through PCI DSS compliant gateways only
- SEC-004: Session management with secure, HTTP-only cookies and CSRF protection
- SEC-005: Comprehensive input validation and sanitisation to prevent injection attacks
- SEC-006: Implementation of OWASP security guidelines and regular vulnerability assessments
- SEC-007: Multi-factor authentication for administrative accounts
- SEC-008: Regular security audits and penetration testing
- SEC-009: GDPR-compliant data handling with right to erasure capabilities

6.3. Reliability Requirements

- REL-001: System uptime shall be 99.5% during business hours
- REL-002: Automatic backup of critical data every 24 hours
- REL-003: Recovery time objective (RTO) of 4 hours
- REL-004: Recovery point objective (RPO) of 1 hour

6.4. Availability Requirements

- AVAIL-001: System shall be available 24/7 with planned maintenance windows
- AVAIL-002: Graceful degradation during partial system failures
- AVAIL-003: Load balancing for high availability

6.5. Maintainability Requirements

- MAINT-001: Modular architecture for easy component updates
- MAINT-002: Comprehensive logging for troubleshooting
- MAINT-003: Code documentation and commenting standards
- MAINT-004: Automated testing coverage minimum 80%

6.6. Portability Requirements

- PORT-001: Database independence through ORM layer
- PORT-002: Cross-platform compatibility
- PORT-003: Containerization support for deployment flexibility

7. Appendices

7.1. Database Schema

Applicants Table

Field	Type	Constraints
applicant_id	SERIAL	PRIMARY KEY
title	VARCHAR(10)	NOT NULL
first_name	VARCHAR(100)	NOT NULL
surname	VARCHAR(100)	NOT NULL
email	VARCHAR(255)	UNIQUE, NOT NULL
mobile_number	VARCHAR(20)	NOT NULL
date_of_birth	DATE	NOT NULL
nationality	VARCHAR(100)	NOT NULL
residency_status	ENUM	'HOME', 'EU', 'INTERNATIONAL'
ucas_id	VARCHAR(20)	UNIQUE
created_at	TIMESTAMPTZ	DEFAULT NOW()
updated_at	TIMESTAMPTZ	DEFAULT NOW()

Table 1: Applicants Table Schema

Verification Table

Field	Type	Constraints
verification_id	SERIAL	PRIMARY KEY
applicant_id	INTEGER	FOREIGN KEY REFERENCES applicants(applicant_id)
verification_token	VARCHAR(255)	NOT NULL
token_type	ENUM	'EMAIL', 'MOBILE', 'DOCUMENT'
expiry_time	TIMESTAMPTZ	NOT NULL
status	ENUM	'PENDING', 'VERIFIED', 'EXPIRED', 'FAILED'
attempts	INTEGER	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()

Table 2: Verification Table Schema

Transactions Table

Field	Type	Constraints
transaction_id	SERIAL	PRIMARY KEY
applicant_id	INTEGER	FOREIGN KEY REFERENCES applicants(applicant_id)
stripe_payment_id	VARCHAR(255)	UNIQUE
paypal_payment_id	VARCHAR(255)	UNIQUE
amount_gbp	DECIMAL(10,2)	NOT NULL
payment_method	ENUM	'STRIPE_CARD', 'STRIPE_DIRECT_DEBIT', 'PAYPAL'
transaction_status	ENUM	'PENDING', 'COMPLETED', 'FAILED', 'REFUNDED'
vat_amount	DECIMAL(8,2)	DEFAULT 0.00
transaction_date	TIMESTAMPTZ	NOT NULL
created_at	TIMESTAMPTZ	DEFAULT NOW()

Table 3: Transactions Table Schema

Programmes Table

Field	Type	Constraints
programme_id	SERIAL	PRIMARY KEY
programme_code	VARCHAR(20)	UNIQUE, NOT NULL
programme_title	VARCHAR(255)	NOT NULL
faculty	VARCHAR(100)	NOT NULL
level	ENUM	'UNDERGRADUATE', 'POSTGRADUATE', 'RESEARCH'
home_fee_gbp	DECIMAL(10,2)	NOT NULL
international_fee_gbp	DECIMAL(10,2)	NOT NULL
application_fee_gbp	DECIMAL(8,2)	DEFAULT 0.00
is_active	BOOLEAN	DEFAULT TRUE

Table 4: Programmes Table Schema

7.2. API Endpoints

Authentication Endpoints

POST /api/v1/auth/register - Applicant registration
POST /api/v1/auth/verify-email - Email verification
POST /api/v1/auth/verify-mobile - Mobile number verification
POST /api/v1/auth/login - Applicant authentication
POST /api/v1/auth/logout - Session termination
POST /api/v1/auth/password-reset - Password reset functionality

Payment Endpoints

GET /api/v1/payment/fees/{applicantId} - Retrieve applicable fees
POST /api/v1/payment/stripe/intent - Create Stripe payment intent
POST /api/v1/payment/paypal/order - Create PayPal payment order
POST /api/v1/payment/webhooks/stripe - Handle Stripe webhooks
POST /api/v1/payment/webhooks/paypal - Handle PayPal webhooks
GET /api/v1/payment/receipt/{transactionId} - Generate payment receipt

Application Endpoints

POST /api/v1/applications/personal-details - Submit personal information
POST /api/v1/applications/qualifications - Submit educational qualifications
POST /api/v1/applications/references - Submit reference details
POST /api/v1/applications/documents/upload - Document upload functionality
GET /api/v1/applications/status/{applicantId} - Application status tracking
POST /api/v1/applications/submit - Final application submission

Administrative Endpoints

GET /api/v1/admin/applications - List all applications (paginated)
GET /api/v1/admin/reports/financial - Financial reporting
PUT /api/v1/admin/applications/{id}/status - Update application status

7.3. User Journey Summary

- Initial Registration: Streamlined account creation with email and mobile verification
- Programme Selection: Browse available courses with clear fee information
- Eligibility Assessment: Automated pre-screening based on entry requirements
- Secure Payment Processing: Multi-gateway payment options with instant confirmation
- Comprehensive Application: Detailed information collection following UCAS standards
- Document Verification: Secure upload and verification of supporting documents
- Application Tracking: Real-time status updates and communication
- Administrative Review: Streamlined processing for admissions staff
- Decision Communication: Automated notification of application outcomes
- Enrolment Integration: Seamless transition to student systems upon acceptance

7.4. Implementation Timeline

Phase	Duration	Completion Status
Requirements Analysis & Design	3 weeks	85%
Backend API Development	6 weeks	75%
Frontend Development	6 weeks	60%
Database Setup & Migration	2 weeks	90%
Payment Gateway Integration	4 weeks	70%
UCAS API Integration	3 weeks	40%
Security Implementation	2 weeks	80%
Testing & Quality Assurance	3 weeks	Planned
User Acceptance Testing	2 weeks	Planned
Production Deployment	1 week	Planned
Post-Launch Support	Ongoing	Planned

Table 5: Implementation Timeline

7.5. Compliance Checklist

- ✓ GDPR Article 13 – Information provided when personal data is obtained
- ✓ GDPR Article 17 – Right to erasure implementation
- ✓ PCI DSS Level 1 compliance certification
- ✓ WCAG 2.1 AA accessibility standards
- ✓ ICO registration and data protection impact assessment
- ✓ University information security policy compliance
- ✓ UCAS technical integration standards adherence

Document Control

Version 1.0

Status Under Review

Next Review Date 15th September 2025

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